Semester project 11
„Spiking neural network for brain anomaly detection“

Topic: Detecting brain anomaly using spiking neural networks
Task: Developing spiking neural networks to classify Electroencephalogram (EEG) segments
Supervisor: Dr. Hongliu Yang, hongliu.yang@tu-dresden.de

Dataset: Intracranial EEG was recorded from dogs with naturally occurring epilepsy. Data are organized in folders containing training and testing data for each canine subject. The training data consist of 1-second EEG clips labeled "Ictal" (seizure segments), or "Interictal" (non-seizure segments). For this study, 80% of the provided training data will be used for model training and the remaining 20% for testing.

Link to dataset: https://www.kaggle.com/competitions/seizure-detection/data

snnTorch: a PyTorch compatible platform for spiking neural networks.

Link to snnTorch: https://snntorch.readthedocs.io/en/latest/

Task description:
1. Read the literature on spiking neural networks [1]
2. Download the dataset and split the data into training and testing sets for each subject. Visualize selected EEG segments to develop an understanding of difference between brain states.
3. Learn to use PyTorch’s Dataset and Dataloader classes to construct your custom EEG dataset.
4. Install snnTorch and experiment with developing spiking neural networks using provided examples.
5. Construct a 3-layer CNN (conv5x8, conv5x16, linear) with LIF neurons. Train it on the MNIST dataset and evaluate its performance. Experiments with different spike encoding methods and loss functions.
6. Construct a 6-layer CNN (conv5x8, conv5x16, 3xconv5x16, linear) with LIF neurons. Train it on the EEG dataset for dog 1 using rate coding and evaluate the performance. Analyze how performances varies with different random initializations.
7. Compare the performance of the spiking neural network with that of the corresponding ANN.
8. **Prepare** your results in a form of presentation slides for the colloquium at the end of the semester.
9. **Comment** your implementation code so that it is understandable even without a detailed report. Add a **Readme file** that explains how to use your implementation code. **Submit** the commented implementation code as one ZIP file to your supervisor **before the examination**.

Literature:

[1] Eshraghian, Jason K., et al. "Training spiking neural networks using lessons from deep learning." *Proceedings of the IEEE* 111.9 (2023): 1016-1054. https://doi.org/10.48550/arXiv.2109.12894